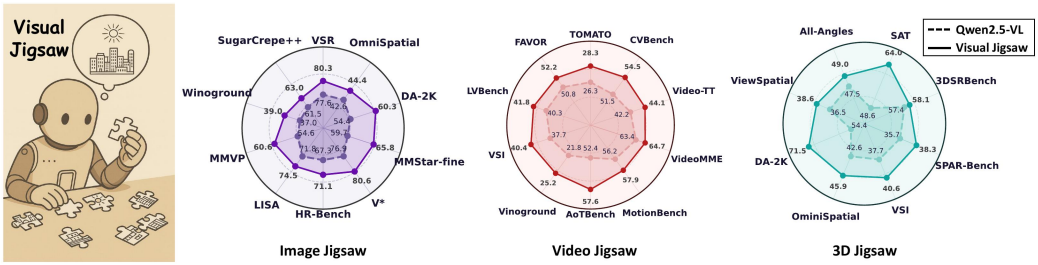

Figure 1: **Visual Jigsaw improves perception across images, videos, and 3D inputs.** The dashed polygon denotes the Qwen2.5-VL baseline and the solid polygon denotes the corresponding Visual Jigsaw model. The largest gains appear on benchmarks that stress fine-grained perception, temporal ordering, and spatial depth understanding.

VISUAL JIGSAW POST-TRAINING IMPROVES MLLMS

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ABSTRACT

Visual Jigsaw is a self-supervised post-training framework for multimodal large language models that targets visual perception directly rather than using images or videos only as sparse evidence for text-centric reasoning. The core idea is simple: partition a visual input into modality-specific elements, shuffle them, and train the model with reinforcement learning from verifiable rewards to recover the correct ordering using text output only. This turns image layout recovery, video chronology recovery, and depth ordering in RGB-D scenes into a unified permutation prediction problem that needs no extra visual decoder, no human annotation, and no architectural change to the base MLLM. We instantiate the approach on images, videos, and 3D data using Qwen2.5-VL-7B-Instruct and GRPO with a graded reward that combines exact-match accuracy, discounted partial correctness, and strict output-format validation. Visual Jigsaw improves fine-grained image understanding, temporal video understanding, and 3D spatial perception simultaneously, raising MMVP from 54.66 to 60.66, AoTBench from 52.41 to 57.64, and SAT-Real from 48.66 to 64.00. Across all three modalities, reinforcement learning transfers better than supervised fine-tuning on the same jigsaw task and preserves reasoning performance when applied to a reasoning-oriented MLLM.

1 INTRODUCTION

Reinforcement learning based post-training has become a strong recipe for improving multimodal large language models (MLLMs), especially on visual reasoning tasks with verifiable answers. Recent examples include Vision-R1, MM-Eureka, ThinkLite-VL, VL-Cogito, LLaVA-Critic-R1, and Video-R1. However, most current post-training objectives remain predominantly text-centric: visual inputs are consumed as supporting evidence for producing better chain-of-thought traces, better answer selection, or better preference judgments. This emphasis has produced impressive progress on multimodal reasoning benchmarks, but it leaves a separate question only partially addressed: how should post-training improve the model's intrinsic visual understanding?

That question matters because many current MLLM failures are perception failures rather than reasoning failures. Recent diagnostics expose weaknesses in fine-grained image perception on benchmarks such as MMVP, MMStar, HR-Bench, and MME-RealWorld, temporal ordering and motion understanding on benchmarks such as AoTBench, Vinoground, Video-MME, TempCom-

pass, TVBench, MotionBench, and CVBench, and 3D spatial reasoning on settings such as SAT, OmniSpatial, ViewSpatial, All-Angles, and 3DSRBench. Several recent vision-centric methods address this by introducing explicit visual reconstruction or proxy perception objectives, including Cambrian, ASVR, ViCrit, and Visual-RFT. While effective, those approaches either add visual-generation style supervision, rely on task-specific verifiers, or focus on narrower visual skills.

We ask whether a general-purpose MLLM can be post-trained on a modality-agnostic, vision-centric task that is simultaneously challenging, self-supervised, and exactly verifiable, while still using text-only outputs. Our answer is Visual Jigsaw. The method converts visual perception into a permutation recovery problem: split the input into image patches, temporal video clips, or depth-ordered points, shuffle them, and ask the model to reconstruct the original order. Because the target is a permutation, the reward is easy to verify; because solving the task requires attending to structure in the visual signal, the supervision is inherently vision-centric.

This formulation has three practical advantages. First, it is plug-and-play for standard text-generating MLLMs, requiring no new output head or visual decoder. Second, it generalizes across image, video, and RGB-D based 3D settings with the same policy-learning interface. Third, it exposes a useful reward-design problem: difficult jigsaw tasks are informative but sparse if trained with exact-match reward only, so we use a graded reward that grants discounted credit for partially correct valid permutations while rejecting invalid outputs.

Our contributions are straightforward:

- We introduce Visual Jigsaw, a self-supervised and verifiable post-training framework that strengthens visual understanding in MLLMs through permutation recovery rather than text-centric reasoning targets.
- We show that the same formulation transfers across images, videos, and 3D depth ordering, improving fine-grained perception, temporal understanding, and spatial reasoning on diverse downstream benchmarks.
- We demonstrate that reinforcement learning transfers better than supervised fine-tuning on the jigsaw task, and that image jigsaw can improve a reasoning-oriented MLLM's visual perception without materially harming its reasoning benchmarks.

2 Related Work

2.1 RL Post-Training for MLLMs

Recent work has extended rule-based reinforcement learning and reinforcement learning from verifiable rewards from language models to multimodal settings. Existing efforts primarily target multimodal reasoning, math, and preference optimization, including Vision-R1, MM-Eureka, ThinkLite-VL, VL-Cogito, LLaVA-Critic-R1, and Video-R1. These methods show that RL can improve multimodal reasoning substantially, but the training signal is usually derived from text-side correctness, reasoning trajectories, or preference judgments. Visual Jigsaw is complementary: instead of using visual content to support better text reasoning, it uses a purely visual ordering objective to improve the model's perceptual grounding.

2.2 Vision-Centric Supervision for MLLMs

Another line of work aims to strengthen visual understanding more directly. Vision-centric architectures and training recipes such as Cambrian and V* emphasize stronger visual grounding and search behavior. Reconstruction-based objectives such as ASVR explicitly supervise visual outputs or semantic visual tokens, while proxy-task formulations such as ViCrit and Visual-RFT define verifiable visual tasks for RL or reward-based fine-tuning. Our method shares the motivation of improving perception directly, but differs in formulation: it does not reconstruct pixels or captions, does not require task-specific external metrics such as IoU, and uses a single ordering abstraction across images, videos, and 3D RGB-D scenes.

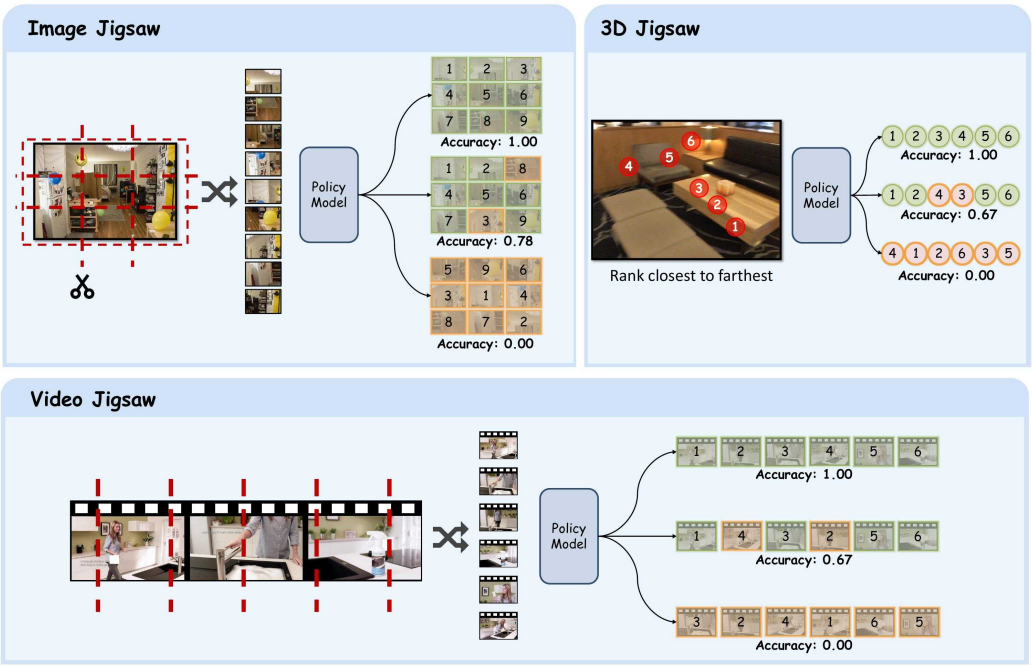

Figure 2: **Illustration of the Visual Jigsaw tasks.** Image Jigsaw shuffles spatial patches and asks for raster-order recovery, Video Jigsaw shuffles temporal clips and asks for chronological recovery, and 3D Jigsaw shuffles depth-ordered points marked on an RGB image and asks for nearest-to-farthest recovery. The same policy-learning interface is used across all three settings.

2.3 Perception Benchmarks

Our evaluations target benchmarks designed to isolate different forms of visual competence. For images, these include fine-grained and compositional benchmarks such as MMVP, MMStar, MM-Bench, HR-Bench, MME-RealWorld, Winoground, and SugarCrepe++; grounding and detection settings including LISA and OVDEval; and spatial perception benchmarks such as VSR, OmniSpatial, and DA-2K. For videos, we evaluate temporal and motion understanding on AoTBench, Vinoground, Video-MME, TempCompass, TVBench, MotionBench, LVBench, Video-TT, and CVBench. For 3D reasoning, we use SAT, 3DSRBench, ViewSpatial, All-Angles, OmniSpatial, VSI-Bench, SPAR-Bench, and DA-2K. These benchmark groups make it possible to test whether a vision-centric post-training task transfers beyond the pretext task itself.

3 Method

Visual Jigsaw formulates vision-centric post-training as a general visual ordering problem. Given a visual input from some modality, we derive a set of K elements using a modality-specific partitioning rule, shuffle them, and ask the policy model to predict the permutation that restores the original structure. The target is emitted as text, so the method works with standard text-only decoders in models such as Qwen2.5-VL-7B-Instruct. Figure 2 shows the three instantiations used in this paper.

3.1 Reward Design

The supervision target is a permutation, which makes verification exact and inexpensive. Let $g = [g_1, \ldots, g_K]$ denote the ground-truth permutation and $o = [o_1, \ldots, o_K]$ the model prediction. A

binary exact-match reward is too sparse for difficult jigsaws, so we use a graded reward:

$$\mathrm{Reward}(o, g) = \begin{cases} 1, & \text{if } o = g, \\ \gamma \cdot \frac{1}{K} \sum_{i=1}^{K} \mathbf{1}[o_i = g_i], & \text{if } \mathrm{ValidPermutation}(o) \wedge o \neq g, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

where $\gamma \in (0, 1)$ discounts partial matches. This design preserves a strong incentive for exact reconstruction while still providing intermediate learning signal for harder tasks. The validity check is critical: any output that is not a proper permutation of size K receives zero accuracy reward, which prevents degenerate strategies such as repeating the same index.

We also enforce a strict response format. The model places its reasoning inside `<think>...</think>` and the final permutation inside `<answer>...</answer>`. Outputs that follow the required format receive an additional format reward of 0.2; invalidly formatted outputs receive zero for both format and accuracy. In all experiments, we set the discount factor for partial correctness to $\gamma = 0.2$.

3.2 Image Jigsaw

For images, we partition an image into a regular $m \times n$ grid of non-overlapping patches and shuffle them. The model must output the indices of the shuffled tiles in the original raster order, from left to right and top to bottom. We use $m = n = 3$, yielding nine tiles per image. Training uses 118K images from COCO, filtering out images whose shorter side is below 84 pixels so that individual tiles remain visually meaningful.

The prompt explicitly lists the nine shuffled tiles and asks the model to recover the original 3×3 arrangement as a comma-separated index sequence. This task forces the model to rely on local texture, object continuity, global layout, and fine-grained inter-patch relations rather than on external labels.

3.3 Video Jigsaw

For videos, we segment a video uniformly along the temporal axis into K non-overlapping clips, shuffle the clips, and ask the model to output their original chronological order. We use $K = 6$ and train on 100K videos from LLaVA-Video. To reduce trivial frame-matching cues, we trim 5% of the frames from both the start and end of each clip. Each clip contains at most 12 frames, and videos shorter than 24 seconds are removed.

The prompt presents the shuffled clips with indices and asks for the chronological permutation. Compared with image jigsaw, this objective emphasizes temporal continuity, motion direction, and event progression. Because the output remains a permutation, the same reward machinery in Eq. 1 applies directly.

3.4 3D Jigsaw

Directly shuffling native 3D primitives would be a natural extension, but current general-purpose MLLMs still consume 3D tasks primarily through 2D images or videos. We therefore design a practical RGB-D variant. Given an RGB-D image, we select $K = 6$ points with distinct depths, sort them from nearest to farthest, and then shuffle the order. The shuffled points are marked on the RGB image with indexed red circles, and the model must recover the nearest-to-farthest order.

Training uses RGB-D data from ScanNet. We generate 300K training samples by randomly selecting six-point combinations under three constraints: points must lie between 0.1 and 10 meters in depth, any two points must be at least 40 pixels apart in the image, and their depth values must differ by more than 0.2 meters. This keeps the ordering task visually resolvable while maintaining substantial variety in scene geometry.

3.5 Training Objective and Practical Setup

We optimize all three tasks with Group Relative Policy Optimization (GRPO). The base model is Qwen2.5-VL-7B-Instruct. In the provided training scripts, we disable KL regularization and entropy

loss, use a learning rate of 1×10^{-6}, and sample 16 responses per prompt. The global batch size is 256 for image jigsaw and 128 for video and 3D jigsaw. The released scripts train the image and video variants for 32 epochs with the above batch sizes and rollout settings; in the corresponding paper configuration, this matches 1000 steps for image and video jigsaw and 800 steps for 3D jigsaw.

4 EXPERIMENTS

4.1 EVALUATION PROTOCOL

We evaluate transfer rather than pretext-task accuracy. Image Jigsaw is tested on fine-grained perception and compositional benchmarks including MMVP, MMStar, MMBench, HR-Bench, V*, MME-RealWorld, LISA-Grounding, OVD-Eval, VSR, OmniSpatial, DA-2K, Winoground, and Sugar-Crepe++. Video Jigsaw is evaluated on AoTBench, Vinoground, TOMATO, FAVOR-Bench, TUNA-Bench, Video-MME, TempCompass, TVBench, MotionBench, LVBench, VSI-Bench, Video-TT, and CVBench. The 3D variant is evaluated on SAT-Real, 3DSRBench, ViewSpatial, All-Angles, OmniSpatial, VSI-Bench, SPAR-Bench, and DA-2K.

We compare against the Qwen2.5-VL-7B baseline and, where available in the provided tables, recent reasoning-oriented or RL-trained MLLMs such as ThinkLite-VL, VL-Cogito, LLaVA-Critic-R1, and Video-R1. All benchmark values are taken directly from the supplied evaluation tables.

4.2 MAIN RESULTS ON IMAGES

Table 1: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
Model	**MMVP**	**MMStar (fine-grained)**	**MMBench**	**HR-Bench-8K**	**V***	**MME-RealWorld**	**LISA-Grounding**	**OVD-Eval**	**VSR**	**OmniSpatial**	**DA-2K**	**Winoground**	**SugarCrepe++**
	test	fine	en_dev	test	test	lite	test	test	test	test	val	g-acc	test
ThinkLite-VL	55.33	59.95	84.19	68.12	76.96	46.17	73.70	35.78	78.09	42.60	58.46	35.25	61.49
VL-Cogito	55.33	56.64	82.98	69.62	79.58	**47.63**	72.26	35.78	79.82	44.29	56.43	38.25	**63.59**
LLaVA-Critic-R1	53.33	57.80	83.16	67.50	78.01	45.18	68.52	35.28	78.50	42.73	53.82	34.75	61.93
Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	**80.68**	43.55	**61.46**	38.75	62.03
Image Jigsaw	**60.66**	**65.81**	**84.45**	**71.13**	**80.63**	45.96	**74.54**	**36.49**	80.36	**44.49**	60.35	**39.00**	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

Table 1 shows consistent gains across all three image benchmark groups. Relative to Qwen2.5-VL-7B, Image Jigsaw improves MMVP by 6.00 points, MMStar-fine by 6.06 points, HR-Bench by 3.75 points, and DA-2K by 5.90 points. The gains extend beyond the most obviously related tasks: we also observe improvements on grounding, open-vocabulary detection, visual spatial reasoning, and compositional matching. These results support the central hypothesis that reconstructing shuffled image structure teaches fine-grained visual correspondences that transfer to diverse image understanding tasks.

4.3 MAIN RESULTS ON VIDEOS

Video Jigsaw improves over the baseline for every reported frame budget and every benchmark in Table 2. At 64 frames, the gains are largest on AoTBench (+5.23), Vinoground (+3.40), and CVBench (+3.00), with additional improvements on Video-MME, MotionBench, VSI-Bench, and Video-TT. This pattern is consistent with the intended supervision: recovering the original clip order

Table 2: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	AoTBench	Vinoground	TOMATO	FAVOR-Bench	TUNA-Bench	VideoMME	TempCompass	TVBench	MotionBench	LVBench	VSI-Bench	Video-TT	CVBench
		vqa	group	test	test	test	wo subs	mc	test	val	test	test	mcq	test
Video-R1	16	45.06	9.40	27.29	49.47	53.00	56.62	70.19	51.80	55.82	34.53	34.34	42.95	47.50
Video-R1	32	47.53	10.20	27.29	49.90	54.26	59.88	71.77	53.54	56.12	38.61	35.11	42.63	48.10
Video-R1	64	48.68	10.60	27.36	50.51	54.33	60.85	72.59	53.43	56.09	38.80	36.61	42.74	48.69
Qwen2.5-VL-7B	16	45.52	12.60	25.87	48.54	53.14	57.44	71.77	49.94	55.56	33.51	32.79	38.39	47.70
Qwen2.5-VL-7B	32	49.48	18.20	26.34	49.34	54.88	60.70	72.59	51.96	56.47	39.19	35.34	41.57	49.60
Qwen2.5-VL-7B	64	52.41	21.80	26.35	50.86	55.79	63.44	72.84	53.74	56.29	40.35	37.74	42.25	51.50
Video Jigsaw	16	51.67	15.20	27.56	49.69	55.10	58.07	73.10	51.33	56.87	36.41	35.39	40.19	49.80
(Gain)		+6.15	+2.60	+1.69	+1.15	+1.96	+0.63	+1.33	+1.39	+1.31	+2.90	+2.60	+1.80	+2.10
Video Jigsaw	32	55.00	21.40	28.03	50.56	56.49	62.37	**73.60**	53.31	**57.99**	39.70	38.47	43.27	51.60
(Gain)		+5.52	+3.20	+1.69	+1.22	+1.61	+1.67	+1.01	+1.35	+1.52	+0.51	+3.13	+1.70	+2.00
Video Jigsaw	64	**57.64**	**25.20**	**28.30**	**52.27**	**56.63**	**64.74**	**73.60**	**54.18**	57.91	**41.83**	**40.40**	**44.11**	**54.50**
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

requires understanding temporal continuity, temporal directionality, and cross-clip event relations, which are exactly the failure modes probed by these benchmarks.

4.4 Main Results on 3D Spatial Understanding

Table 3: **Evaluation results on 3D benchmarks.** 3D Jigsaw consistently enhances performance on both directly related depth comparison tasks (DA-2K) and broader 3D perception tasks spanning single-view, multi-view, and egocentric video inputs.

Model	SAT-Real	3DSRBench	ViewSpatial	All-Angles	OmniSpatial	VSI-Bench	SPARBench	DA-2K
	test	test	test	test	test	test	tiny	test
Qwen2.5-VL-7B	48.66	57.42	36.52	47.56	42.66	37.74	35.75	54.45
3D Jigsaw	**64.00**	**58.13**	**38.62**	**49.06**	**45.99**	**40.64**	**38.31**	**71.56**
(Gain)	+15.34	+0.71	+2.10	+1.50	+3.33	+2.90	+2.56	+17.11

Table 3 shows especially strong transfer for the RGB-D based 3D variant. The direct depth-related benchmark DA-2K improves by 17.11 points, and SAT-Real improves by 15.34 points. We also observe gains on broader spatial reasoning settings, including single-view 3D reasoning, multi-view reasoning, and egocentric video spatial understanding. These improvements indicate that the task is not merely teaching a narrow ordering heuristic; it appears to strengthen the model's ability to interpret spatial structure from visual evidence.

5 Analysis

5.1 RL Versus SFT on the Same Jigsaw Task

The image benchmark table also includes an Image Jigsaw (SFT) model trained on the same task with supervised fine-tuning. While SFT provides moderate gains on some benchmarks, it is consistently weaker than RL and can even hurt transfer. For example, SFT lowers LISA-Grounding from 71.89 to 66.59 and OVD-Eval from 35.07 to 34.35, whereas RL raises them to 74.54 and 36.49. SFT

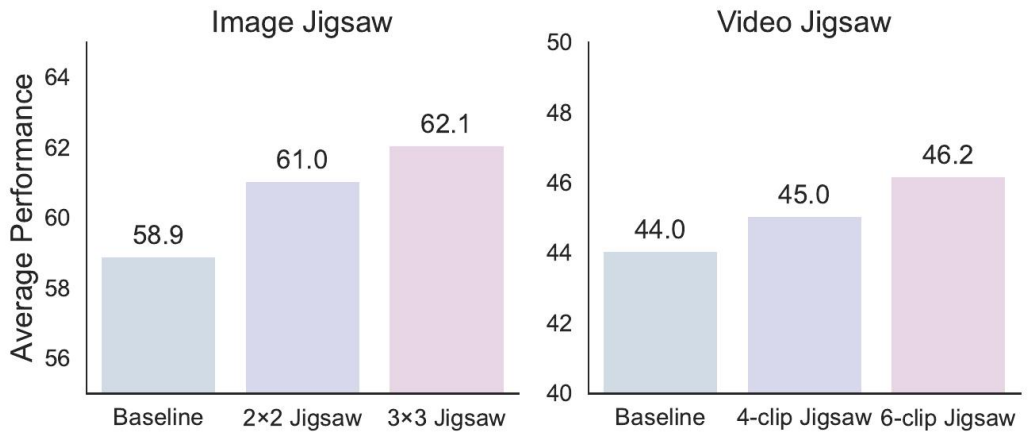

Figure 3: Performance with different jigsaw difficulties on image and video tasks. Harder tasks provide stronger downstream gains: the average image score rises from 58.9 for the baseline to 61.0 with a 2×2 jigsaw and 62.1 with a 3×3 jigsaw, while the average video score rises from 44.0 to 45.0 and then 46.2 as the task increases from four clips to six clips.

does achieve the best score on a few columns, including VSR and DA-2K, but the broader pattern favors RL as the more reliable way to turn a vision-centric proxy task into transferable perceptual skill.

5.2 Effect of Jigsaw Difficulty

Figure 3 studies how pretext difficulty affects transfer. Easier variants still help, but they help less. On images, moving from the baseline to a 2×2 jigsaw raises the average score from 58.9 to 61.0, while the full 3×3 jigsaw reaches 62.1. On videos, the average rises from 44.0 for the baseline to 45.0 with a four-clip jigsaw and 46.2 with the six-clip task. The trend suggests that more demanding reconstruction problems expose richer supervisory signal, provided that the reward remains informative enough for learning.

5.3 Applying Visual Jigsaw to a Reasoning Model

Table 4: Performance of Visual Jigsaw on a reasoning-oriented MLLM (ThinkLite-VL), showing improved visual perception while preserving reasoning ability.

Model	Vision-Centric Avg	MathVista testmini	MathVision testmini	MathVerse testmini	MMMU val	EMMA mini
ThinkLite-VL	59.69	**75.20**	30.92	**50.76**	**55.11**	26.75
+ Image Jigsaw	**61.60**	75.10	**35.20**	50.50	54.22	**29.00**

Table 4 evaluates whether Image Jigsaw can improve a reasoning-oriented model rather than only a perception-oriented base model. Starting from ThinkLite-VL, the image jigsaw stage raises the vision-centric average from 59.69 to 61.60. The reasoning benchmarks remain broadly stable, with gains on MathVision and EMMA and small declines on MathVista, MathVerse, and MMMU. The result is important because it shows that a vision-centric RL stage is not redundant with reasoning-oriented RL and can be layered on top of it to recover perceptual gains that text-centric post-training may leave under-optimized.

5.4 Why the Same Objective Transfers Across Modalities

Although the three tasks use different visual inputs, they share the same supervision pattern: the model must infer structure that is present in the observation but hidden by a permutation.

- Image jigsaw rewards spatial continuity and object-boundary consistency.
- Video jigsaw rewards temporal continuity, motion direction, and event progression.
- 3D jigsaw rewards relative depth comparison and scene-layout consistency.

These common constraints explain why transfer is strongest on benchmarks that stress boundaries, motion, and depth: the policy is trained to reconstruct global organization from local evidence rather than only emit a task-specific format.

Selective transfer. A second pattern is that gains are selective rather than uniform. Image Jigsaw improves MMVP, MMStar-fine, and HR-Bench more than MMBench or SugarCrepe++, which is consistent with a task that sharpens local correspondences and part-whole structure more than broad semantic priors. Video Jigsaw shows the same behavior: AoTBench and Vinoground gain more than TempCompass or TVBench, suggesting that clip reordering mainly strengthens temporal direction and continuity. In 3D, the largest gains appear on SAT-Real and DA-2K, the two benchmarks most directly tied to relative depth comparison. This selectivity argues against a generic formatting effect and supports the interpretation that the learned improvement is genuinely perceptual.

6 CONCLUSION

Visual Jigsaw reframes image, video, and RGB-D perception as a unified ordering problem that can be optimized with reinforcement learning from verifiable rewards. The formulation is simple, self-supervised, and compatible with standard text-only MLLM outputs. Across the supplied experiments, it improves fine-grained image understanding, temporal video understanding, and 3D spatial reasoning without requiring architectural changes or visual-generation components. The results also clarify two broader lessons: first, vision-centric post-training can deliver benefits that are complementary to reasoning-centric RL; second, for hard permutation tasks, reward design matters, and graded partial-credit rewards make these tasks practical for post-training. The strongest gains occur where downstream evaluation depends on recovering latent structure from local evidence, matching the inductive bias of the jigsaw objective.

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